

Aditya Kumar

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Research Interests

Computational and quantum physics applied to quantum hardware: superconducting qubits and circuit QED, spintronics, and the simulation-driven design of quantum chips and materials.

Education

Indian Institute of Technology (Indian School of Mines), Dhanbad Jul 2026 – Jun 2028 (expected)
M.Sc. Physics Dhanbad, India

University of Delhi, Acharya Narendra Dev College Oct 2020 – May 2023
B.Sc. (Hons.) Physics • CGPA 7.608/10 New Delhi, India

Don & Donna Convent School (CBSE) 2015 – 2019
Class XII: 88.88% (Physics, Chemistry, Mathematics) • Class X: CGPA 10/10, 1st rank in batch India

Research Experience

Research Project: Superconducting Quantum Chip Design (Circuit QED) Aug 2026 – Present
Supervisor: Asst. Prof. Rahul Gupta IIT (ISM), Dhanbad

- Designing a superconducting quantum chip layout built around microwave resonators for circuit QED, modelled and simulated in COMSOL Multiphysics.
- Iterating on resonator geometry through simulation to improve the Q-factor, reducing photon loss to support longer-lived, higher-fidelity qubit readout.

Research Intern: Quantum Simulator Development (Source Code) May 2026 – Jun 2026
Supervisor: Prof. Pinaki Majumdar Remote

- Built a Python quantum simulator solving the Schrödinger equation for the harmonic oscillator and quartic double-well potential, validating numerical eigenvalues and wave functions against analytic results.
- Simulated quantum tunnelling by time-evolving a Gaussian wave packet and tracking its spreading and barrier transmission, released open-source as a reusable tool for studying bound states and tunnelling.

Lab Technician, Virtual Lab Jan 2022 – 2022
Acharya Narendra Dev College, University of Delhi New Delhi, India

- Developed pages of the college Virtual Lab website (in collaboration with IIT Delhi) using HTML and CSS.
- Designed computer-science experiments with step-by-step guidance, enabling first-time users to perform lab exercises remotely without supervision.

Increasing Solar Panel Efficiency with Water Cooling Jul 2020 – Oct 2021
Group project, SPIE Student Chapter Delhi, India

- Contributed to a group experiment testing water cooling as a method to counter heat-driven efficiency loss in photovoltaic panels.
- Compared panel output with and without cooling to assess water cooling as a low-cost way to raise solar power yield.

Technical Skills

Programming: Python (NumPy, SciPy, Matplotlib), C++, Bash, SciLab

Scientific tools: L^AT_EX, Jupyter, Git/GitHub, Markdown, Quarto

Video and design: DaVinci Resolve, Adobe Premiere Pro, Adobe After Effects, Canva, Tinkercad

Systems and hardware: Linux (Ubuntu, Fedora), Windows, macOS; Arduino UNO R3, 8085 microprocessor